3Laws

Release beta

Thomas Gurriet (tgurriet@3laws.io)

Nov 02, 2023

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Chapter 1

Introduction

Here is the public documentation for 3LawsRobotics Products:

- Robot Diagnostic Module (Robot Diagnostic Module):

Chapter 2

Robot Diagnostic Module

The **Robot Diagnostic Module (RDM) of 3LawsRobotics** is a powerful tool to gather analytics about your robot. It provides low level metrics such as CPU usage, signal delays but also high level safety critical metrics like safety envelope violation, planning inconsistency or sensor damage detection.

All of these metrics can be viewed on your web browser at 3laws.app.

In order to get your account setup and use freely this RDM, please contact contact@3lawsrobotics.com

This product is still at a Beta development stage. New features are developed every month and some breaking changes can still happened. The 3LawsRobotics Team is available to answer your question.

For now only a ROS1, ROS2 interface is available but we intend to extend the offer of possible interfaces in the near future with support for MavLink or MQTT for instance.

2.1 Installation

The Robot Diagnostic Module is distributed as a debian package. A package is available for each of the following configuration

Ubuntu Distribution	ROS1 version	ROS2 version	Architecture
22.04	N/A	Humble/Iron	amd64/arm64v8
20.04	Noetic	Galactic/Foxy	amd64/arm64v8
18.04	Melodic	N/A	amd64/arm64v8

Here is the one liner that will download a public script that will, based on your system download the right package.

```
sudo bash -c "$(wget -qO - https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh)"
```

If you want to install a specific package in a non interactive mode, here is the process:

- Download the script:
  `wget https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh`
- Make it executable
  `chmod +x install.sh`
- Run it with your arguments
  `sudo ./install.sh [-hyf] [-r <ROS_DISTRO>] [-a <ARCH>] [-v <UBUNTU_VERSION>]` if `-yf -r <ROS_DISTRO> -a <ARCH> -v <UBUNTU_VERSION>` specified, the script is non interactive

2.2 Launch the RDM

To launch the RDM, you only have to source your ros distribution: `source /opt/ros/<DISTRO>/setup.sh` and to run the command `ros2 launch lll_rdm rdm.launch.py`

You can add the following command line options:

- use_sim_time: <Boolean, default: false> Specify if the RDM use the ros time or the machine time
- config_filename:
  <String> Path to the config file if not in the `/opt/3lawsRoboticsInc/robot_diagnostic_module/config` folder (**needs to start with /**) or just the config file name inside the default folder.
- robot_id: <String> name of your robot
- log_level: <trace | debug | info | error, default: info> Log level of the RDM
- dry_run: <Boolean> Start the RDM, check the config file and turn off without error if everything has been launched correctly

2.3 Configuration

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The configuration is written inside a YAML. file. This file can be stored anywhere but the default folder is `/opt/3lawsRoboticsInc/robot_diagnostic_module/config`

Other path can be specified in the command line launch command

The Robot Diagnostic Module is shipped with a configuration template that can be found here
`/opt/3lawsRoboticsInc/robot_diagnostic_module/config/template_config.yaml`

In order to have the more user friendly configuration experience, a lot of the fields are optional. The configuration can be build alongside your robot so that it does not represent a burden for the developer.

2.3.1 General configuration rules

Bounds

All bounds parameters are similar:

- norm_type: Norm to use for norm bound, possible values (none, L_1, L_2, L_infinity)
- norm_upper_bound: Upper norm bound, floating point
- norm_lower_bound: Lower norm bound, floating point
- upper_bounds: Upper bounds on the value of the components of the tracking_error, sequence of floating point
- lower_bounds: Lower bounds on the value of the components of the tracking_error, sequence of floating point
- rates_upper_bounds: Upper bounds on the value of the components of the tracking_error, sequence of floating point
- rates_lower_bounds: Lower bounds on the value of the components of the tracking_error,

sequence of floating point

Set both (rate_)upper_bounds[i] and (rates_)lower_bounds[i] to .inf to disable bound on component i.

Leave empty to deactivate all (rate) bounds checking.

Both (rate_)upper_bounds and (rates_)lower_bounds must have same size

Duration parameters

Parameters like signal_min_rate represent durations. They can be specified by an integer number corresponding to a number of nanoseconds:

signal_min_rate: 1 -> 1 nanoseconds

signal_min_rate: 1000000000 -> 1 second

They can also be specified by an integer of floating point followed by the time scale:

signal_min_rate: 1ns -> 1 nanoseconds

signal_min_rate: 25us -> 25 microseconds

signal_min_rate: 0.5ms -> 500 milliseconds

signal_min_rate: 1s -> 1 second

signal_min_rate: 20hz -> 50 milliseconds

signal_min_rate: 1min -> 1 minute

signal_min_rate: 1h -> 1 hour

signal_min_rate: 1d -> 1 day

signal_min_rate: 1y -> 1 year

2.3.2 Scheme

The configuration has **4** distinct parts

- credentials/credentials (Credentials):
  This part is here to specify the credentials to access the databases. These credentials are provided to you by email once your 3LawsRobotics account is created
- diagnostic_config/diagnostic_config (Diagnostic Configuration):
  This part is the more important one. It describe your robot desired behavior, the interface data stream access for the RDM to monitor.
- interface/interface (Interface):
  Here are specified the diagnostic specific options such has the criticality of a certain event or the threshold above which a incident is flagged
- robot_description/robot_description (Robot Description):
  And this last part allow the user to specify interface specific options such has ROS Quality of Service